

AI Model Providers Comparison & Transformer Concepts

Home Assignment

Part 1: AI Model Providers Released in the Last 12 Months

This section compares 10 AI model providers based on release date, knowledge cutoff, token pricing, context window, model type, and primary use cases.

Provider	Model	Release	Knowledge Cutoff	Cost	Context	Type	Best Use
OpenAI	GPT-5	Aug 2025	Varies	Paid API	Up to 1M	Closed	General AI, coding
Anthropic	Claude Sonnet 4.5	Sep 2025	Varies	Paid API	Up to 1M	Closed	Enterprise AI
Google DeepMind	Gemini 3 Pro	Nov 2025	Varies	Paid API	Up to 1M	Closed	Search & multimodal
xAI	Grok 4	Jul 2025	Varies	Paid API	Large	Closed	Web reasoning
Meta	Llama 4 Maverick	Apr 2025	Varies	Free/Open	Large	Open	Fine-tuning
Mistral AI	Magistral Medium	Jun 2025	Varies	Paid API	Large	Open/Commercial	Coding
Alibaba Cloud	Qwen 3.8-Max	Aug 2026	Varies	Paid API	1M	Open	RAG
DeepSeek	DeepSeek V4-Flash	Jul/Aug 2026	Varies	Low Cost	1M	Open	Chatbots
Moonshot AI	Kimi K3	Jul 2026	Varies	Low Cost	1M	Open	Research
NVIDIA	Cosmos Predict 2.5	2026	N/A	Enterprise	Large	Open	Robotics

Part 2: Transformer Concepts

Self-Attention Mechanism

Self-attention enables each word to focus on the most relevant words in the sentence, allowing the model to understand context and long-range relationships.

Example: Example: In 'The animal didn't cross the street because it was tired,' the word 'it' refers to 'animal'.

Key Points

- Identifies important relationships.
- Creates context-aware representations.
- Improves language understanding.

Vector Embedding

Vector embeddings convert words into numerical vectors that capture semantic meaning. Similar words have similar vector representations.

Example: Example: 'King' and 'Queen' have similar embeddings.

Key Points

- Converts text into numbers.
- Captures semantic similarity.
- Input to the Transformer.

Feed Forward Neural Network (FFN)

The FFN processes each token after self-attention, refining its representation and learning more complex patterns.

Example: Example: It refines whether 'bank' means a financial institution or a river bank.

Key Points

- Refines contextual features.
- Applies non-linear transformations.
- Improves prediction accuracy.

Overall Transformer Workflow

1. Vector Embedding
2. Self-Attention
3. Feed Forward Neural Network
4. Repeat across multiple Transformer layers
5. Generate final prediction